Green Labs HR Analytics: Employee Monitoring and Attrition Analysis

Power BI project report | Muneeza Syed

Tools: Power BI, DAX, Power Query, Kimball dimensional modelling (snowflake schema)

1. Executive Summary

Green Labs is a technology company whose HR team needed a single, reliable place to monitor workforce metrics and to understand why employees leave. An end-to-end Power BI solution is designed and built covering requirements gathering, data transformation, dimensional modelling, DAX measure development, and a four-page interactive report.

The dashboard shows a workforce of **1,470 employees**, of whom **1,233 are active** and **237 have left**, giving an overall **attrition rate of 16.1%**. Technology employs two thirds of active staff, and the Attrition page lets HR break the rate down by role, hire year, tenure, business travel and overtime so that retention efforts can be targeted rather than generic.

Headline numbers

Metric	Value
Total employees	1,470
Active employees	1,233
Inactive employees (left)	237
Attrition rate	16.1%
Age range	18 to 51
Largest department (active)	Technology, 828 (67.15%)

2. Business Context and Objectives

Primary objective: Give the HR team a monitoring tool for key employee metrics (headcount, demographics, performance and satisfaction).

Secondary objective: Identify the factors associated with employee attrition so HR can act on them.

Questions the report answers

1. How many employees does Green Labs have, and how is that split across departments and job roles?
2. Who are our employees (age, gender, marital status, ethnicity, pay)?
3. How do individual employees rate their job, environment, relationships and work-life balance over time, and how are they rated by managers?
4. Which groups, roles and working conditions show the highest attrition?

3. Approach

The project followed two phases.

Step 1: Foundations

- Requirements gathering with the HR stakeholder need in mind
- Connecting to the data sources
- Data transformation and cleaning in Power Query
- Building the data model
- Writing the initial DAX measures

Step 2: Presentation

- Branding (consistent Green Labs look and feel)
- Defining the report layout and navigation
- Building the report with chart visualisations

4. Data Model

The **Kimball dimensional modelling** approach was followed, separating measurable events (facts) from descriptive context (dimensions: the who, what, when, where and why). The final model is a **snowflake schema**.

Fact table

- **FactPerformanceRating** stores employees' yearly performance reviews. It has 11 columns and multiple rows per employee, and it is the central point of the model. Columns include EmployeeID, PerformanceID, ReviewDate, EnvironmentSatisfaction, JobSatisfaction, RelationshipSatisfaction, ManagerRating, SelfRating and TrainingOpportunitiesTaken.

Dimension tables

Table	Purpose
DimEmployee	Employee attributes: age and age bins, attrition flag, business travel, department, distance from home (km), education, education field and more
DimEducationLevel	Lookup for education level; snowflaked off DimEmployee
DimDate	Custom-built calendar table with year, quarter, month and day attributes
DimRatingLevel	Translates rating IDs 1 to 5 into labels (Unacceptable, Needs Improvement, Meets Expectation, Exceeds Expectation, Above and Beyond)
DimSatisfiedLevel	Translates satisfaction IDs 1 to 5 into labels (Very Dissatisfied to Very Satisfied)

A dedicated **_Measures** table holds all DAX measures so they are easy to find and keep the field list tidy.

Modelling decisions worth highlighting

- **Reusable lookup tables:** the rating and satisfaction dimensions are each shared by several fact columns (for example, several satisfaction questions all map to one DimSatisfiedLevel). This avoids duplicating labels and keeps the model compact.
- **Two date relationships:** DimDate connects to the fact table on review date and also to DimEmployee (shown as a secondary relationship in the model view), which supports analysis by review period as well as by hire date. This is what the "Date" variants of the measures rely on.
- **Snowflaking education:** education level sits in its own table related to DimEmployee, normalising a low-cardinality attribute.

5. DAX Measures

Measures are grouped by purpose.

Group	Measures	Purpose
Headcount	TotalEmployees, ActiveEmployees, InactiveEmployees	Core workforce counts
Date-aware headcount	TotalEmployeesDate, InactiveEmployeesDate	Same counts evaluated over the hire-date relationship for trend charts
Attrition	% Attrition Rate, % Attrition Rate Date	Share of employees who left, overall and by hire year
Compensation	AverageSalary	Pay comparison, for example by ethnicity
Performance and satisfaction	JobSatisfaction, EnvironmentSatisfaction, RelationshipSatisfaction, WorkLifeBalance, ManagerRating, SelfRating	Average 1 to 5 scores over time and per employee
Review schedule	LastReviewDate, NextReviewDate	Show an individual's review cadence on the Performance Tracker

The attrition rate follows the standard definition: inactive employees divided by total employees (237 / 1,470 = 16.1%).

6. Dashboard Walkthrough and Findings

The report has four pages sharing consistent branding and a navigation bar.

6.1 Overview

- KPI cards: **1,470** total, **1,233** active, **237** inactive, **16.1%** attrition.
- **Employee Hiring Trends:** a 100% stacked column of hires by year (2012 to 2022) split by attrition status, showing how long-term retention differs by hiring cohort.
- **Active employees by department:** Technology **828 (67.15%)**, Sales **354 (28.71%)**, Human Resources the remaining 51 (about 4.1%).
- **Department and job role breakdown:** Technology is made up of roles such as Software Engineer, Data Scientist, Machine Learning, Senior Software Engineer, Engineering Manager and Analytics Manager; Sales includes Sales Executive, Sales Representative and Manager.

Takeaway: Green Labs is a technology-led business, so retention in engineering and data roles has an outsized effect on delivery capacity.

6.2 Demographics

- Employees range from **18 to 51** years old, grouped into <20, 20-29, 30-39, 40-49 and 50+ bins.
- **Age and gender:** a 100% stacked chart shows the gender mix inside each age band (Female, Male, Non-Binary, Prefer Not To Say), useful for spotting representation gaps.
- **Marital status:** Married **624 (42.45%)**, Single **549 (37.35%)**, Divorced **297 (20.2%)**.
- **Ethnicity and average salary:** headcount and average salary side by side for each ethnic group, with salaries clustering between roughly $100K and $120K. This view supports pay-equity checks.

Takeaway: the demographic view gives HR the baseline for diversity and fair-pay monitoring. Any ethnic group whose average salary sits notably below the rest should be reviewed against role and tenure before conclusions are drawn.

6.3 Performance Tracker

An employee-level page with a search and select list. Choosing a person shows their **start date, last review and next review**, plus six year-by-year trend charts (2016 to 2022) on a 1 to 5 scale:

- Job Satisfaction
- Relationship Satisfaction
- Self Rating
- Work Life Balance
- Manager Rating
- Environment Satisfaction

Reference legends explain the rating scale (1 Unacceptable to 5 Above and Beyond) and satisfaction scale (1 Very Dissatisfied to 5 Very Satisfied).

Takeaway: managers can prepare for reviews quickly, compare **self rating against manager rating**, and spot a falling satisfaction trend before it becomes a resignation.

6.4 Attrition

The diagnostic page, built to answer the secondary objective. It opens with the **16.1%** attrition rate and breaks it down by:

Visual	Question answered
Attrition by department and job role	Which roles lose the most people?
Attrition by hire date	Do certain hiring years churn more?
Attrition by business travel (frequent, some, none)	Does travel burden drive people out?
Attrition by overtime requirement (yes / no)	Is overtime associated with leaving?
Attrition by tenure (years at company, 0 to 10)	When in the employee lifecycle do people leave?

7. Business Recommendations

The recommendations below follow from the dashboard's structure and should be confirmed against the exact values on the Attrition page before presenting.

1. **Target retention at the highest-attrition roles.** Use the department and job role view to rank roles by attrition rate and set role-specific retention plans (compensation review, career path, workload) rather than company-wide programmes.
2. **Review overtime and travel policies.** If overtime and frequent travel show elevated attrition, introduce overtime caps, time-off in lieu, travel rotation or a travel allowance for the affected groups.
3. **Strengthen the critical early-tenure window.** Where attrition clusters at a given tenure band, add structured onboarding, a 90-day check-in and a mentor, and run stay interviews around the tenure point where leaving peaks.
4. **Use the Performance Tracker as an early warning system.** Flag employees whose job satisfaction, work-life balance or manager rating trends downward over successive reviews, and trigger a manager conversation.
5. **Close self-rating and manager-rating gaps.** Large gaps indicate misaligned expectations. Calibrate review criteria and train managers in feedback conversations.
6. **Run a pay-equity review.** Use the ethnicity and salary view to compare like-for-like roles and tenures, and address any unexplained differences.
7. **Review hiring cohorts.** If particular hire years show weaker retention, investigate what differed (recruitment channel, market conditions, onboarding) and apply the lessons to current hiring.
8. **Invest in development.** Use the TrainingOpportunitiesTaken field to test whether training participation relates to satisfaction and retention, and widen access if it does.

8. Limitations and Next Steps

Limitations

- Attrition factors are shown as descriptive associations; they do not prove causation.
- Results are shown one factor at a time. Combined effects (for example, overtime among frequent travellers) are not yet isolated.
- The dataset is a snapshot, so exit reasons and compensation changes are not captured.

Next steps

- Add exit-interview reason codes and compensation history to the model.
- Build a predictive attrition model (for example logistic regression or gradient boosting in Python) to score flight risk.
- Add a drill-through page from the Attrition view to the Performance Tracker.
- Publish to Power BI Service with row-level security and a scheduled refresh.

9. Skills Demonstrated

- **Requirements gathering** and translating a business problem into KPIs
- **Data transformation** with Power Query
- **Dimensional modelling:** Kimball method, fact and dimension design, snowflake schema, shared lookup dimensions, multiple date relationships
- **DAX:** counts, rates, date-aware measures and a dedicated measures table
- **Report design:** branding, navigation, KPI cards, trends and drill-down
- **Data storytelling:** turning charts into actionable business recommendations